

Online Examination Report

Date:	12.08.2014
Attempts:	1
Examinee:	Muhanguzi, Allan
Date of birth:	2/10/1988
Subject:	040-Human performance
Result:	55 %
Time used:	00:30:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	174	40.	HUMAN PERFORMANCE	1,00	1,00
2	164	40.	HUMAN PERFORMANCE	1,00	1,00
3	175	40.	HUMAN PERFORMANCE	0,00	1,00
4	179	40.	HUMAN PERFORMANCE	0,00	1,00
5	197	40.	HUMAN PERFORMANCE	0,00	1,00
6	168	40.	HUMAN PERFORMANCE	1,00	1,00
7	181	40.	HUMAN PERFORMANCE	1,00	1,00
8	178	40.	HUMAN PERFORMANCE	1,00	1,00
9	196	40.	HUMAN PERFORMANCE	1,00	1,00
10	206	40.	HUMAN PERFORMANCE	1,00	1,00
11	78	40.1.3	Flight safety concepts	1,00	1,00
12	68	40.2.1.1	The atmosphere	0,00	1,00
13	16	40.2.1.1	The atmosphere	1,00	1,00
14	71	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
15	60	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
16	63	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
17	67	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
18	49	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
19	10	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
20	47	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
21	64	40.2.2	Man and Environment: the sensory system	1,00	1,00
22	76	40.2.2.1	Central, peripheral and autonomic nervous system	1,00	1,00
23	86	40.2.2.2	Vision	0,00	1,00
24	13	40.2.2.2	Vision	0,00	1,00
25	58	40.2.2.5	Integration of sensory inputs	0,00	1,00
26	8	40.2.2.5	Integration of sensory inputs	1,00	1,00
27	30	40.2.3	Health and hygiene	0,00	1,00
28	87	40.2.3.4	Intoxication	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	153	40.3	BASIC AVIATION PSYCHOLOGY	0,00	1,00
30	65	40.3.1.4	Response selection	1,00	1,00
31	81	40.3.3.1	Decision-making concepts	1,00	1,00
32	5	40.3.4.3	Co-operation	0,00	1,00
33	59	40.3.4.4	Communication	0,00	1,00
34	128	40.3.5	Human behaviour	0,00	1,00
35	113	40.3.5	Human behaviour	0,00	1,00
36	123	40.3.5	Human behaviour	0,00	1,00
37	11	40.3.5.1	Personality, attitude and behaviour	1,00	1,00
38	1	40.3.5.1	Personality, attitude and behaviour	1,00	1,00
39	48	40.3.6.2	Stress	1,00	1,00
40	44	40.3.6.2	Stress	0,00	1,00
Total:55%				22,00	40,00

Attachment 2: List of result by topic

Licence

Date: 12.08.2014

Examinee: Muhanguzi, Allan

Location: Uganda Civil Aviation Authority

Your Result: 22,00 from 40,00 Points (55,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
40. HUMAN PERFORMANCE	40	22,00	40,00	55,00
40.1. Human Factors: basic concepts	1	1,00	1,00	100,00
40.2. Basic aviation physiology and health maintenance	17	9,00	17,00	52,94
40.3. BASIC AVIATION PSYCHOLOGY	12	5,00	12,00	41,67
Total	40	22,00	40,00	55,00

1

One of the steps in the aeronautical decision making (ADM) process involved in good decision making is

- ☒ identifying personal attitudes hazardous to safe flight.
- ☐ making a rational evaluation of the required actions.
- ☐ developing a 'can do' attitude.

2

Rapid or extra deep breathing while using oxygen can cause

- ☒ **hyperventilation.**
- ☐ **a build-up of carbon dioxide in the body.**
- ☐ **cyanosis.**

3

Examples of classic behavioral traps that experienced pilots may fall into are to

- ☐ **complete a flight as planned, please passengers, meet schedules, and 'get the job done.'**
- ☒ **assume additional responsibilities and assert PIC authority.**
- ☐ **promote situational awareness and then necessary changes in behavior.**

4

Hazardous attitudes which contribute to poor pilot judgment can be effectively counteracted by

- ☐ an appropriate antidote.
- ☒ early recognition of these hazardous attitudes.
- ☐ taking meaningful steps to be more assertive with attitudes.

5

An abrupt change from climb to straight-and-level flight can create the illusion of

☒ a descent with the wings level.

☐ a noseup attitude.

☐ tumbling backwards.

6

What suggestion could you make to people who are experiencing motion sickness?

- ☐ Have the students lower their head, shut their eyes, and take deep breaths.
- ☐ Recommend taking medication to prevent motion sickness.
- ☒ Avoid unnecessary head movement and focus sight on a point outside the aircraft.

7

In the aeronautical decision making (ADM) process, what is the first step in neutralizing a hazardous attitude?

☐ Recognizing the invulnerability of the situation.

☐ Making a rational judgement.

☒ Recognizing hazardous thoughts.

8

Name some hazardous attitudes that can affect your judgment during the aeronautical decision making (ADM) process.

☐ Peer pressure and stress levels.

☒ Antiauthority, impulsivity, and resignation.

☐ Impulsivity, antiestablishment, and reevaluation.

9

A sloping cloud formation, an obscured horizon, and a dark scene spread with ground lights and stars can create an illusion known as

☐ elevator illusions.

☒ false horizons.

☐ autokinesis.

10

What action should be taken if hyperventilation is suspected?

- ☐ Breathe at a slower rate by taking very deep breaths.
- ☐ Consciously force yourself to take deep breaths and breathe at a faster rate than normal.
- ☒ Consciously breathe at a slower rate than normal.

11

A pilot using a checklist is an example of the interaction within the SHELL Concept of:

☐ L and L

☐ L and E

☒ L and S

☐ L and H

12

Fatigue and stress

- ☐ increase the tolerance to hypoxia
- ☐ will increase the tolerance to hypoxia when flying below 15 000 feet
- ☒ do not affect hypoxia at all
- ☐ lower the tolerance to hypoxia

13

Which gas below 70,000 ft., above ground gas makes up the major part of the atmosphere ?

☐ Carbon dioxide

☐ Oxygen

☒ Nitrogen

☐ Ozone

14

When exhaling, the expired air contains:

- ☐ more nitrogen than in the inhaled air
- ☐ more oxygen than in the inhaled air
- ☐ less water vapour than in the inhaled air
- ☒ more carbon dioxide than in the inhaled air

15

The following actions are appropriate when faced with symptoms of decompression sickness:

1. climb to higher level
2. descent to the higher of 10000 ft or MSA and land as soon as possible
3. breathe 100 % oxygen
4. obtain medical advice about recompression after landing

☐ 1, 2 and 3 are correct

☐ 1 and 3 are correct

☒ 2, 3 and 4 are correct

☐ 1 and 4 are correct

16

When flying above 10.000 feet hypoxia arises because:

- ☒ percentage of oxygen is lower than at sea level
- ☐ partial oxygen pressure is below the critical value of 55mm/Hg
- ☐ composition of the air is different from sea level
- ☐ composition of the blood changes

17

Flying immediately following a dive with SCUBA diving equipment to a depth greater than 10 metres

- ☐ will always lead to hypoxia
- ☒ can cause decompression sickness even when flying at pressure altitudes below 18 000 FT
- ☐ has no influence on altitude flights
- ☐ presents any dangers caused by DCS (decompression sickness) when climbing to altitudes not exceeding 30 000 FT

18

Which of the following statements are correct:

- 1: SCUBA diving may be practiced without restriction
- 2: Many medicines have effects which are incompatible with flight safety
- 3: An adequate amount of fluid should be drunk when flying
- 4: Alcohol has no effect on the inner ear.

☐ 1, 3 and 4 are correct

☒ 2 and 3 are correct

☐ 2, 3 and 4 are correct

☐ 1, 2 and 3 are correct

19

The rate and depth of breathing is primarily regulated by the concentration of:

- ☐ nitrogen in the air
- ☐ oxygen in the cells
- ☒ carbon dioxide in the blood
- ☐ water vapour in the alveoli

20

The respiratory process consists mainly of

- ☐ the transportation of oxygen to the cells and the elimination of nitrogen
- ☒ the transportation of oxygen to the cells and the elimination of carbon monoxide
- ☐ the diffusion of oxygen through the respiratory membranes into the blood, transportation to the cells, diffusion into the cells and elimination of carbon dioxide from the body
- ☐ the transportation of carbon dioxide to the cells and elimination of oxygen

21

Perceptual illusions are:

- ☐ usually not serious except in bad weather.
- ☐ never caused by the brain's ability to distort normal sense inputs.
- ☐ abnormal and indicate weakness in one or more of the senses.
- ☒ normal and can be prevented by trusting instrument read-out.

22

A stereotype and involuntary reaction of the organism on stimulation of receptors is called:

☐ **change of stimulation level**

☒ **reflex**

☐ **control system**

☐ **data processing**

23

The retina of the eye

- ☐ is the muscle, changing the size of the crystalline lens
- ☐ filters the UV-light
- ☐ is the light-sensitive inner lining of the eye containing the photoreceptors essential for vision
- ☒ only regulates the light that falls into the eye

24

When focussing on near objects:

- ☒ the shape of lens gets more spherical
- ☐ the cornea gets smaller
- ☒ the pupil gets larger
- ☐ the shape of lens gets flatter

25

What would be the effect if, in a tight turn, one bends down to pick up a pencil?

☐ Coriolis effect.

☒ Vertigo.

☐ Inversion Illusion.

☐ Barotrauma.

26

Which sensations does a pilot get, when he is rolling out of a prolonged level turn?

- ☐ Flying straight and level
- ☐ Turning into the original direction
- ☒ Turning in the opposite direction
- ☐ Climbing

27

The following can be observed when the internal body temperature falls below 35°C:

- ☐ the appearance of intense shivering
- ☒ profuse sweating
- ☐ mental disorders, and even coma
- ☐ shivering, will tend to cease, and be followed by the onset of apathy

28

Which statement is correct?

1. Smokers have a greater chance of suffering from coronary heart disease
2. Smoking tobacco will raise the individuals physiological altitude during flight
3. Smokers have a greater chance of contracting lung cancer

☐ 2 and 3 are correct, 1 is false

☐ 1,2 and 3 are correct

☒ 1 and 3 are correct, 2 is false

☐ 1 and 2 are correct, 3 is false

29

Probably the greatest single barrier to effective communication is the

- ☒ **use of inaccurate statements.**
- ☐ **lack of a common core of experience between communicator and receiver.**
- ☐ **use of abstractions by the communicator.**

30

The ability to monitor information which could indicate the development of a critical situation

- ☐ is responsible for the development of inadequate mental models of the real world
- ☒ is necessary to maintain good situational awareness
- ☐ makes no sense because the human information processing system is limited anyway
- ☐ is dangerous, because it distracts attention from flying the aircraft

31

Judgement is based upon:

- ☐ the development of skills through constant practice of flight manoeuvres
- ☐ a decision-making process involving the 5 physical senses and their use to manually operate the aircraft controls
- ☒ a process involving a pilot's attitude to take and to evaluate risks by assessing the situation and making decisions based upon knowledge, skill and experience
- ☐ the ability to interpret the flight instruments

32

Which of the following statements best characterises a self-centered cockpit ?

- ☒ The egoistic and self-centered personality of the captain often leads to a synergetic cockpit
- ☐ While decreasing communication, the independence of each member bolsters the crew's synergy
- ☐ Each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on
- ☐ The communication between crew members always increases when the captain takes charge of a situation

33

"Stereotypes" are preconceptions or prejudices which can lead us to:

- ☐ develop better teamwork by standardizing procedures.
- ☒ act in the same manner in all situations and thus assuring stability.
- ☐ communicate non-verbally with a stranger.
- ☐ mis-judge individuals even if we have contact with them.

34

A pilot and friends are going to fly to an out-of-town football game. When the passengers arrive, the pilot determines that they will be over the maximum gross weight for takeoff with the existing fuel load. Which of the following alternatives best illustrates the RESIGNATION reaction?

- ☐ Weight and balance is a formality forced on pilots by the FAA.
- ☐ Well, nobody told him about the extra weight.
- ☒ He can't wait around to de-fuel, they have to get there on time.

35

The basic drive for a pilot to demonstrate the 'right stuff' can have an adverse effect on safety, by

- ☐ generating tendencies that lead to practices that are dangerous, often illegal, and may lead to a mishap.
- ☒ a total disregard for any alternative course of action.
- ☐ imposing a realistic assessment of piloting skills under stressful conditions.

36

The passengers for a charter flight have arrived almost an hour late for a flight that requires a reservation. Which of the following alternatives best illustrates the ANTIAUTHORITY reaction?

- ☐ If the pilot hurries, he or she may still make it on time.
- ☒ The pilot can't help it that the passengers are late.
- ☐ Those reservation rules do not apply to this flight.

37

For a normal and healthy person, personality traits are:

- ☐ easy changeable.
- ☐ unstable.
- ☒ stable.
- ☐ easy changed by an outside influence.

38

Attitudes are defined as:

- ☐ the conditions necessary for carrying out an activity
- ☐ a synonym for behaviour
- ☒ tendencies to respond to people, things or events in a particular manner
- ☐ the genetic predispositions for thinking and acting

39

The behavioural effects of stress may include:

- 1: manifestation of aggressiveness.
- 2: a willingness to improve communication.
- 3: a willingness for group cohesion.
- 4: a tendency to withdrawal.

The combination of correct statements is:

- ☐ 3 and 4 are correct
- ☐ 2,3 and 4 are correct
- ☒ 1 and 4 are correct
- ☐ 1,2 and 3 are correct

40

When can the General Adaption Syndrome(GAS) activate?

- ☐ Responding to an imaginary threat.
- ☒ Responding to boredom.
- ☐ Responding to Hypovigilance.
- ☐ Responding to Circadian Disrhythmia.